Elementary Particle as a Closed Wave of Field 01

Scalar Profile, Normal-Retention Interpretation, Mass, Memory, and the Absence of a Point Singularity

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Abstract

Field 01 explores whether a particle-like object can be represented by a localized closed phase configuration rather than by a point core. Its interpretive vocabulary links circulation, scalar-profile retention, and memory. The mathematical comparison layer uses scalar and gauge profiles with vortex-like winding; it does not establish a stable three-dimensional elementary particle. In particular, the present benchmark does not derive a physical photon or a transition from an open mode to a closed configuration. A separate fixed-background numerical report supplies a restricted profile benchmark, not a particle mass or a stability theorem. This paper states the localization, regularity, dynamical, and observational questions that must be resolved before the interpretation can become a physical model. The proposed horizon connection remains a separate hypothesis.

Keywords: Field 01; closed wave; elementary particle; phase circulation; scalar profile; VEV-like profile; normal-retention interpretation; mass; memory; quantum field theory; horizon; holography.

1 How to Read This Paper

This paper is a working research text, not a completed physical theory. Its purpose is to formulate selected ideas of Field 01 in a language that can be discussed, criticized, and compared with established physics.

The reader should distinguish three levels of statements:

1. **Established physics**: quantum fields, vacuum correlations, photon energy $E = \hbar\omega$, mass–energy equivalence, the stress–energy tensor, and horizon entropy.
2. **Interpretation proposed by Field 01**: a particle is described as a stable field regime, a photon as open phase transport, and a massive particle as a closed node with a local scalar / VEV-like profile that receives a normal-retention interpretation in Field 01.
3. **Open hypothesis**: the scalar / VEV-like profile, its normal-retention interpretation, memory, phase tension, horizon surface recording, and the non-literal evaporation picture used in Field 01 require further mathematical formalization.

Therefore, all formulas not derived from a strict action principle or from a standard theory should be read as working notation and as directions for future formalization. Field 01 does not claim that the Standard Model is wrong, nor does it replace quantum field theory. It offers an additional phase-topological language whose consistency must be tested.

2 Status of This Paper in the Research Program

At the present checkpoint, the particle paper is the most developed paper in the Field 01 public archive. Other directions—cosmology, horizon interpretation, gravitational language, information, and possible extensions toward Standard Model language—remain in progress and should not be read as completed conclusions.

This order is methodological. If the minimal object cannot be defined carefully—a particle as a stable field regime—then larger constructions lose their basis. The central test question of the present version is therefore not the whole Field 01 program, but the narrower question:

can an elementary particle be described as a stable closed phase configuration of a field?

The public repository and open discussions are intended to test this question. Supportive interpretations are welcome, but criticism is equally important: mathematical contradictions, comparison with known soliton and topological models, incompatibility with quantum field theory, missing references, and suggestions for an action, energy functional, or stability condition.

This paper should therefore be read as a working hypothesis and mathematical comparison, not as a completed particle model. Its claims are stated so they can be checked mathematically and compared with established field-theoretic constructions.

3 Aim of the Paper

Field 01 proposes to treat physical structures not as a collection of ready-made objects in a pre-given space, but as stable regimes of a single field. This paper isolates the minimal core of the model: a possible description of an elementary particle as a closed wave.

The main aim is to introduce a working vocabulary that can connect the intuitive description with a mathematical framework. The proposed formalization is not presented as a replacement for the Standard Model. It is an interpretive and hypothetical layer that may help organize questions about particles, mass, memory, and horizons.

The central thesis is:

> An elementary particle can be interpreted as a stable closed phase configuration of Field 01, in which the phase circulation is self-consistent and a local scalar / VEV-like profile is retained; the language of normal retention is an interpretation of that scalar-profile layer, not an already established new standard degree of freedom.

4 Two Regimes of Field 01

Let $\mathcal{F}_{01}$ denote the field considered in the model. At the qualitative level it is described by two elementary regimes:

- 0 — fixation, holding, boundary, recording;
- 1 — direction, motion, circulation, phase transfer.

These symbols should not be read as binary digits in an ordinary computational sense. They are minimal names for two complementary modes of field behavior. The regime 1 expresses directional phase motion, while the regime 0 expresses the ability of the field to hold, close, or record a configuration.

A physical process in Field 01 is not the motion of an object through an empty background. It is a change of regime in the field itself. The basic rhythm can be written schematically as

$$0 \leftrightarrow 1. \tag{1}$$

This expression is not yet an equation of motion. It is a compact notation for the alternation between fixation and directed phase change.

5 Phase and Circulation

To describe a wave-like regime we introduce a phase variable

$$\varphi = \varphi(x, t), \tag{2}$$

where x denotes position in an effective space and t denotes an effective time parameter. At this stage both space and time are treated as descriptive variables, not as independently explained fundamental entities of the model.

A phase circulation along a contour $\mathcal{C}$ may be written as

$$\oint_{\mathcal{C}} d\varphi. \tag{3}$$

If the circulation does not close, the phase is transported without forming a localized node. If the circulation becomes self-consistent, a closed wave configuration can arise. A minimal topological condition for such a configuration is

$$\oint_{\mathcal{C}} d\varphi = 2\pi n, \qquad n \in \mathbb{Z}. \tag{4}$$

This condition is not meant to define a complete particle spectrum. It only expresses the idea that a stable node requires a quantized or self-consistent phase return.

6 Minimal Mathematical Setup

To move from image to testable structure, it is useful to write a minimal set of variables. Let the local field state be approximated by an amplitude and a phase,

$$\psi(x) = A(x)e^{i\varphi(x)}. \tag{5}$$

For a closed node it is also useful to consider two phase branches,

$$\psi_a(x) = A_a(x)e^{i\varphi_a(x)}, \qquad a = 1, 2, \tag{6}$$

and their local separation,

$$\Delta\varphi(x) = \varphi_1(x) - \varphi_2(x). \tag{7}$$

These two branches are not two substances. They are a way to describe internal self-consistency of circulation: the same wave structure must be able to return to itself after a complete phase cycle.

In this notation a closed particle is not a point. It is a map from a circulation contour into a state space,

$$\Gamma : S^1 \longrightarrow \mathcal{S}_{01}, \tag{8}$$

where S^1 denotes the closed circulation parameter and $\mathcal{S}_{01}$ is the still-to-be-defined state space of Field 01. The winding number can then be written as

$$n = \frac{1}{2\pi} \oint_{\mathcal{C}} d\varphi. \tag{9}$$

For physical interpretation, winding alone is not enough. At least four conditions are required:

1. **closure**: observable quantities return to themselves after a full circulation;
2. **localization**: the node energy is concentrated in a finite region and approaches a vacuum-like background outside it;
3. **regularity**: the energy density does not contain a point infinity;
4. **stability**: a small deformation does not immediately destroy the node or lower its energy without a barrier.

These conditions are the first mathematical filter for the model. If they cannot be realized in an explicit energy functional or action, then "particle as a closed wave" remains only a metaphor. If they can be realized, the next question is whether the resulting stable classes can be compared with known particle masses, spins, and charges.

7 Photon as Open Circulation

In standard physics a photon is a quantum of the electromagnetic field, with energy

$$E = \hbar\omega. \tag{10}$$

At the interpretation level, Field 01 associates the photon with an open mode 1. This means transport of direction, phase, and energy without a localized closed node. The current benchmark uses a complex scalar and an Abelian–Higgs-like vortex; it does not yet realize this identification.

If the field is written as

$$\psi = Ne^{i\varphi}, \tag{11}$$

then at $N = 0$ the phase φ is undefined and the phase kinetic term proportional to $N^2(\partial\varphi - A)^2$ vanishes. Therefore "photon = $N = 0$" does not describe a propagating scalar phase wave. In a locally broken $U(1)$ model, the scalar phase is not an independent massless mode either; it belongs to the massive vector sector.

A safer schematic statement is

$$\text{open mode } 1: \qquad \mathcal{R}_0 = 0, \qquad \text{physical representative not yet defined,} \tag{12}$$

where $\mathcal{R}_0 = 0$ means absence of a localized profile deformation, not a vanishing scalar modulus. A photon comparison must separately reproduce gauge invariance, masslessness, two polarizations, and spin 1.

8 Elementary Particle as a Closed Wave

An elementary particle is interpreted as a stable closed configuration of the field. Unlike a photon, it is not merely a transfer of phase along an open direction. It is a self-returning phase regime.

The schematic condition for such a regime is

$$\oint_{\mathcal{C}} d\varphi = 2\pi n. \tag{13}$$

Here $\mathcal{C}$ does not have to be understood as a material wire or a literal circular path. It denotes the effective phase cycle that returns the field to a self-consistent state.

In this interpretation the particle is not a small ball and not a mathematical point with an infinite core. It is a localized mode of field rhythm. Closure alone does not establish stability; the latter requires an explicit energy functional and a perturbation analysis.

9 Scalar Profile and the Interpretation of Local Depth

A standard vortex benchmark contains a scalar modulus N with vacuum value N_0. The profile satisfies $N \to N_0$ far from the node and usually $N \to 0$ at the vortex core. Therefore the bare value of N cannot be identified directly with mode 0 or with an amount of retention.

A localized diagnostic may instead measure departure from the vacuum profile:

$$\mathcal{R}_0(x) = \frac{\left(N^2(x) - N_0^2\right)^2}{N_0^4}, \qquad \mathcal{N} \equiv \text{interpretation}(\mathcal{R}_0). \tag{14}$$

This quantity is gauge-invariant and vanishes on the vacuum background, but it is only a diagnostic, not a fundamental definition of mode 0. The localized structure is represented by the combined phase, scalar, and gauge profile.

If the closed node unfolds, the cautious limit is

$$\mathcal{R}_0 \to 0, \qquad (N, \varphi, A_\mu) \to \text{vacuum or open sector}, \tag{15}$$

rather than the mandatory condition $N \to 0$.

10 Localization and Absence of a Point Singularity

An important requirement for a particle as a closed wave is finite energy. If $\varepsilon(x)$ denotes an effective energy density, a localized node should satisfy

$$E_{\text{node}} = \int_{\mathbb{R}^3} (\varepsilon(x) - \varepsilon_{\text{vac}})\, d^3x < \infty. \tag{16}$$

Here ε_{vac} denotes the background relative to which the configuration energy is measured. This does not define the final Lagrangian, but it fixes an obligatory requirement: particle mass should not arise from infinite self-energy concentrated at a point center.

In a spherically symmetric approximation, regularity of the core can be expressed as

$$\varepsilon(0) < \infty, \qquad \int_0^{r_0} 4\pi r^2 \varepsilon(r)\, dr < \infty. \tag{17}$$

If the local scalar / VEV-like profile is written as $N(r)$, natural regular boundary conditions are

$$N(0) = N_c < \infty, \qquad \partial_r N(0) = 0, \tag{18}$$

while outside the node one may require

$$N(r) \to N_{\text{vac}} \quad \text{or} \quad N(r) \to 0, \tag{19}$$

depending on the future formalization.

In this form, absence of a point singularity becomes a mathematical condition rather than only a philosophical thesis. The model must produce finite energy, a smooth or weakly singular field configuration, and no need to place mass at a point of zero size. If a chosen action does not satisfy these conditions, that version of the model must be rejected or changed.

11 Mass as Energy of Retaining the Scalar Profile

In standard physics mass is related to energy by

$$E = mc^2, \tag{20}$$

and, in the Standard Model, elementary particle masses are connected with their coupling to the Higgs field. Field 01 does not deny this structure. It asks whether mass can also be described in an additional phase-geometric language.

In the model, mass is interpreted as the energy cost of maintaining a closed phase configuration and its local scalar / VEV-like profile. A schematic expression is

$$mc^2 \sim E_{\text{closed}}[\varphi, N, A_\mu]. \tag{21}$$

A possible functional form is

$$E_{\text{closed}} = \int_{\Sigma} \left(A|\nabla\varphi|^2 + B|N|^2 + V(\varphi, N) \right) d\Sigma, \tag{22}$$

where A, B, and V are not yet fixed from a fundamental action. This schematic expression is not a gauge-invariant action for the local vortex benchmark. A physical energy functional must specify covariant derivatives, gauge-field energy, normalization, and boundary conditions. The intended comparison is with the rest energy of a localized solution, not with a mass already calculated here.

In words:

> Mass is associated with the energy required to hold a closed phase rhythm and prevent it from unfolding into an open mode.

12 Stability of the Closed Node

The next level of testing is stability. A closed phase circulation must not only be drawn; it must arise as a stationary configuration of some energy functional. In minimal form this can be written as

$$\delta E[A, \varphi, N] = 0. \tag{23}$$

Stationarity alone is not enough: a maximum or saddle point does not define a stable particle. One therefore needs at least local non-negativity of the second variation within an allowed class of deformations,

$$\delta^2 E[\eta] \geq 0, \tag{24}$$

where η denotes a small deformation of amplitude, phase, contour, or local profile.

The topological class acts as a restriction on allowed deformations. If the winding number n is conserved, the node cannot continuously become vacuum without a discontinuity, profile suppression, or departure from the chosen state class:

$$n \neq 0 \quad \Rightarrow \quad \Gamma \not\sim \Gamma_{\text{vac}} \tag{25}$$

under fixed boundary conditions. This does not prove the existence of a particle, but it specifies the form of a future proof: one must find an action in which such classes exist and have finite energy.

A working test of the model should therefore proceed in this order:

1. choose the field space and boundary conditions;
2. construct an energy functional or action;
3. find stationary closed solutions;
4. check finite energy and absence of a point core;
5. check the spectrum of small perturbations around the solution;
6. compare stable classes with quantum numbers.

Until these steps are performed, Field 01 remains a research program. The value of the program is that the hypothesis can be criticized in a concrete way: either the listed conditions can be realized, or they cannot.

13 A Dynamical Core Instead of a Point Core

The usual problem of a point-like particle is that if all structure is concentrated at a mathematical point, singular behavior may appear in the description. Field 01 replaces the point core by a dynamical core.

The core of the particle is not a point at which quantities must diverge. It is the closed self-consistency of the phase circulation. The particle is localized not because it contains a hard center, but because its phase rhythm returns to itself.

This allows the model to avoid an internal point singularity at the level of interpretation. It does not by itself solve all ultraviolet problems of quantum field theory. Rather, it suggests a different geometric intuition: elementary localization should be described by stable closed field modes rather than by structureless points.

14 Memory as an Interphase Structure

Field 01 uses the term memory not as psychological memory and not as a hidden classical record inside a particle. Memory means the retained pattern of phase relations that makes a configuration identifiable and dynamically continuous.

Let

$$\mathcal{M} = \mathcal{M}[\varphi, \mathcal{N}, \mathcal{C}] \tag{26}$$

denote the information retained by a closed configuration. This notation is schematic. It means that memory is carried not by a material shape alone, but by phase relations, circulation, and the stability of the local scalar / VEV-like profile.

This point is important for the later discussion of horizons. If memory is phase-structural rather than object-like, then a limiting surface can record the phase pattern without transporting a solid object through the horizon.

15 Forces as Regimes of a Closed Wave

The Field 01 interpretation describes interactions in terms of how closed waves coordinate, hold, exchange, or open their phase structures. In this paper this idea is kept as a qualitative guide rather than as a completed derivation of the known interactions.

A cautious mapping is the following:

- electromagnetic interaction may be associated with phase asymmetry and open phase exchange;
- strong interaction may be associated with tight coordination of local scalar profiles / VEV-profile layers and local phase locking;
- weak interaction may be associated with instability, reconfiguration, or decay of a closed phase regime;

- gravity may be associated with collective deformation of the field and with the energetic cost of stable configurations.

This mapping is not a substitute for gauge theory. The lines concerning strong and weak interaction require special caution: in established physics they are described by specific gauge symmetries and fields. Field 01 must either derive such structures as effective descriptions or clearly state the boundary of its applicability.

It is a research program: the next step would be to identify whether the phase variables of Field 01 can reproduce gauge symmetries, charges, and known conservation laws.

16 Relation to Established Physics

Quantum Field Theory

Quantum field theory already treats particles as excitations of fields. Field 01 is compatible with this broad idea, but proposes a specific interpretation: a massive particle is not merely any excitation, but a closed and self-consistent phase regime.

The model should therefore be compared with solitons, topological defects, bound states, and other stable field configurations. The analogy is useful, but not yet a derivation.

Vacuum Correlations

The vacuum in quantum field theory is not empty. It contains correlations and fluctuations. Field 01 uses this as a conceptual bridge: what appears as a particle may be treated as a stable pattern inside a deeper field structure.

No claim is made here that Field 01 has already derived the quantum vacuum. The claim is more modest: the language of phase structure and closure is not alien to modern field-based physics.

Spin

Spin is a quantum number with precise transformation properties under rotations and Lorentz symmetry. Field 01 does not yet derive spin. It suggests that spin may eventually be related to the topology of phase circulation and to the way a closed wave returns to itself.

A future formalization would have to show how half-integer and integer spin representations arise. Until then, the relation between spin and closed circulation remains a hypothesis.

Higgs Mechanism

The Standard Model explains elementary masses through coupling to the Higgs field. Field 01 does not replace this mechanism at the present stage. A possible bridge is to interpret mass as the energy of maintaining a stable closed configuration, while the Higgs mechanism supplies the established field-theoretic description of mass generation.

A serious comparison would require deriving effective mass terms from a Field 01 action and checking whether they can be related to known Higgs couplings.

Stress–Energy Tensor

In general relativity the stress–energy tensor describes the distribution of energy and momentum that sources spacetime curvature. Field 01 may eventually require its own effective stress–energy tensor for phase configurations.

At the current stage, however, expressions such as phase tension or scalar-profile retention energy are only working terms. They should not be confused with an already defined tensorial theory.

17 Horizon Interpretation

The following schematic limit belongs to the earlier horizon hypothesis. It is not derived from the vortex equations or from the profile diagnostic $\mathcal{R}_0$, and does not establish an evaporation law.

The same logic that distinguishes an open photon mode from a closed massive particle also suggests a limiting regime near a horizon. If the local scalar / VEV-like profile is suppressed,

$$\mathcal{N} \to 0, \tag{27}$$

then the localized bulk-like structure can no longer be described as an ordinary closed particle in volume.

In Field 01 the horizon is interpreted as a recording surface. The closed configuration does not have to be imagined as mechanically torn apart. Rather, its phase memory can be transferred into a surface code:

$$\mathcal{M}_{\text{bulk}} \longrightarrow \mathcal{M}_{\text{boundary}}. \tag{28}$$

This is intentionally close to the general spirit of the holographic principle and to the Bekenstein–Hawking relation between entropy and horizon area. It is not presented as a derivation of black-hole thermodynamics. It is a model-specific interpretation of why information may be associated with a boundary rather than with an internal object-like volume.

18 Current Numerical Status

A separate numerical report records the first restricted checkpoint for a two-component profile on a fixed background. It studies difference and aligned localizers within a preselected parametric family. The retained internal diagnostic is

$$\xi_{\text{br,max}} = 2.0934591793773114 \times 10^{-3}. \tag{29}$$

The connected-branch fold was checked at three matching radii, $R = 30$, 40, and 50, with a maximum relative spread of 0.074515%. Nine profile coordinates were then re-audited around the retained point. The largest remaining predicted gain is 0.917518%, below the predeclared one-percent threshold for authorizing another direct fold calculation.

This checkpoint establishes only internal reproducibility within the restricted fixed-background ansatz. It does not derive an elementary particle, a fundamental action, a physical mass, a spectrum of quantum numbers, or dynamical stability. The numerical report is therefore a technical benchmark, not evidence that Field 01 is a physical theory. Its public consistency script checks the supplied summaries; it does not rerun the underlying boundary-value calculations.

19 Program for Further Formalization

To turn Field 01 into a mathematical theory, several tasks are necessary.

State Space

One must define the state space of the field:

$$\Psi_{01} \in \mathcal{H}_{01} \tag{30}$$

or an equivalent classical field configuration space. It must be clear what the fundamental variables are and how physical states are identified.

Action or Energy Functional

A dynamical principle is required. A possible schematic action is

$$S_{01} = \int \mathcal{L}(\varphi, \partial_\mu \varphi, \mathcal{N}, \partial_\mu \mathcal{N})\, d^4x. \tag{31}$$

The Lagrangian must be chosen or derived so that stable closed configurations exist.

Topological Charge

Closed waves should be classified by a topological quantity, for example

$$Q = \frac{1}{2\pi} \oint_{\mathcal{C}} d\varphi. \tag{32}$$

A stable particle-like solution would correspond to a non-trivial value of Q.

Stability Condition

A particle should correspond to a stable or metastable extremum of the energy:

$$\delta E_{\text{closed}} = 0, \qquad \delta^2 E_{\text{closed}} > 0 \tag{33}$$

within an appropriate class of variations.

Bulk-to-Boundary Transition

The transition toward a horizon should be described by a limiting map

$$\Pi_\partial : \mathcal{M}_{\text{bulk}} \to \mathcal{M}_{\text{boundary}}, \tag{34}$$

with the local scalar / VEV-like profile suppressed:

$$\mathcal{N} \to 0. \tag{35}$$

Comparison with Known Quantum Numbers

Finally, the model must explain or reproduce known quantum numbers: charge, spin, mass, flavor, and gauge representation. Without this, Field 01 remains an interpretive framework rather than a predictive physical theory.

20 What Is Claimed and What Remains Hypothetical

The paper makes the following limited claims:

- it is possible to formulate a coherent phase-topological vocabulary for Field 01;
- an elementary particle can be interpreted as a closed phase configuration;
- the photon is provisionally associated with an open mode; the present vortex benchmark does not derive its physical representative;
- mass can be associated with the energetic cost of maintaining a closed configuration;
- a horizon can be interpreted as a limiting surface where the local scalar / VEV-like profile is suppressed and phase memory becomes boundary record.

The following points remain open:

- derivation from a precise action;
- reproduction of the Standard Model particle spectrum;
- derivation of spin and gauge charges;
- quantitative relation to the Higgs mechanism;
- construction of an effective stress–energy tensor;
- testable predictions that differ from existing theories.

Thus Field 01 should currently be read as a structured research program, not as a completed theory of nature.

21 Questions for Public Discussion

For open technical discussion and later preprint work, the next mathematical iteration should answer questions such as:

1. Can one construct a simple energy functional that admits finite closed phase nodes?
2. Is one phase field enough, or is a multicomponent field, spinor structure, or gauge field required from the beginning?
3. Can the local scalar / VEV-like profile N be defined without adding an unnecessary independent entity?
4. Do stable solutions with non-trivial winding and finite energy exist in three spatial dimensions?
5. Can twisted circulation be related to fermionic 4π periodicity without a mechanical rotation picture?
6. Can mass as the energy of a stable solution be compared with known mass terms and the Higgs mechanism?
7. What observable or theoretical distinction could separate Field 01 from a pure reformulation of known quantum field theory?

These questions are deliberately formulated so that a negative answer is possible. If mathematical analysis shows that such closed phase nodes are impossible or unstable, that is important information for the model. If a minimal stable example is found, the next task is to compare its spectrum with established physics.

22 Conclusion

Field 01 proposes a closed phase configuration as a possible description of a particle-like object. Winding and scalar-profile retention provide a language for this hypothesis, but neither establishes localization, stability, physical mass, charge, or spin.

The present scalar/gauge benchmark describes a closed-sector comparison. It does not identify the photon with a vanishing scalar modulus or derive a photon-to-particle transition. The separate fixed-background numerical checkpoint remains a restricted calculation, while the horizon interpretation remains a hypothesis requiring its own dynamical analysis.

The next substantive task is to specify the dynamics and observables needed to distinguish this proposal from the standard field-theoretic constructions on which it draws.